statsreasoning6

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---

1. Generalized linear models

```r
library(tidyverse) # For data wrangling
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.0     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.2     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.1
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becom
```

```r
library(brms) # For stats
```

```
Loading required package: Rcpp
Loading 'brms' package (version 2.23.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').

Attaching package: 'brms'
```

```
The following object is masked from 'package:stats':

    ar
```

```r
library(ggeffects) # for plotting model predictions
# Note: I needed to also install the `insight` and `see` packages to get `modelbased` to inst
# install.packages('modelbased') # if you need to install this package
library(modelbased) # for plotting model predictions. supports the link scale (ggeffects does
```

```
Attaching package: 'modelbased'

The following objects are masked from 'package:ggeffects':

    collapse_by_group, pool_predictions, residualize_over_grid
```

```r
# install.packages('faraway') # if you need to install this package
library(faraway) # For data on galapagos species richess
```

```
Attaching package: 'faraway'

The following object is masked from 'package:brms':

    epilepsy
```

Conceptual practice

Q1.1a What values can each of the response variables take on?

1. Non negative integers
2. 0/1
3. Real numbers between 0 and 100
4. Real numbers
5. Non negative real numbers

Q1.1b Choose a distribution that fits each of the response variables

1. Poisson
2. Bernoulli
3. Beta
4. Gaussian
5. Gamma

Q1.2 Choose a distribution that fits your final project response variable

Response variable: Abundance (counts) of harbor seals or stellar sea lions at Ano Nuevo State Park

Values: Non negative integers (whole numbers)

Distribution: Poission

1.2 GLM with a log link

Explore the data

```r
# Read in the pre-stored data
data("gala")
# Check out the first 6 rows
head(gala)
```

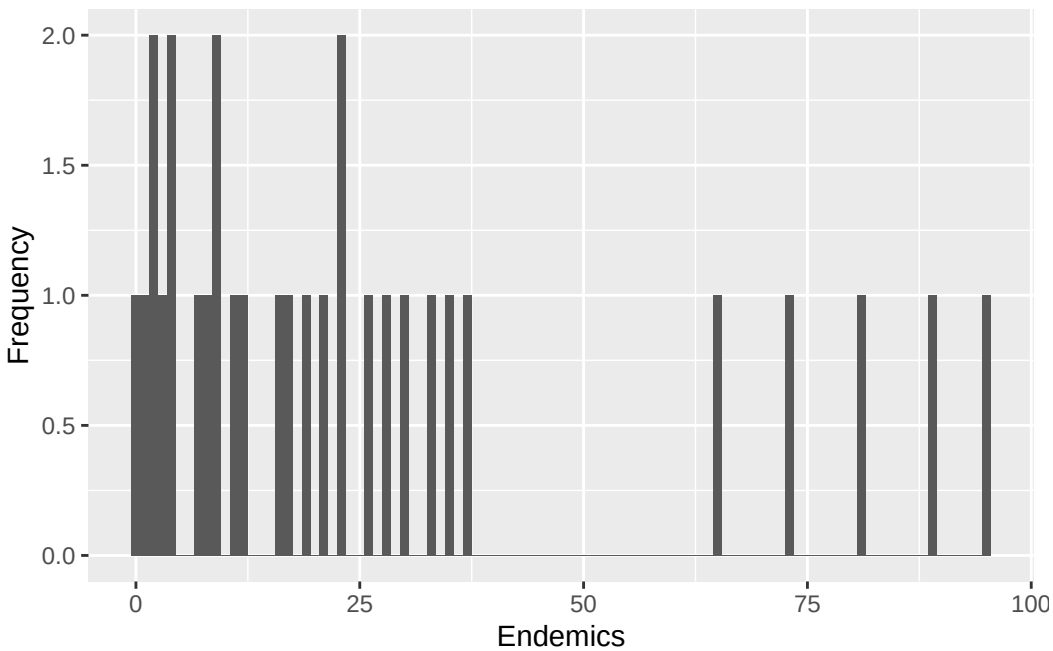
```
             Species Endemics  Area Elevation Nearest Scruz Adjacent
Baltra            58       23 25.09       346     0.6   0.6     1.84
Bartolome         31       21  1.24       109     0.6  26.3   572.33
Caldwell           3        3  0.21       114     2.8  58.7     0.78
Champion          25        9  0.10        46     1.9  47.4     0.18
Coamano            2        1  0.05        77     1.9   1.9   903.82
Daphne.Major      18       11  0.34       119     8.0   8.0     1.84
```

```r
# Look at the help page too!
```

Q1.3 Plot a histogram of the response variable `Endemics`

Plot a histogram or density plot of the response variable `Endemics` to see how it is distributed. Does it look like a normal (Gaussian) distribution, which is a nice, symmetrical distribution that can take any value, or does it have certain constraints? What are those constraints?

```
ggplot(gala,aes(x=Endemics)) +
  geom_histogram(binwidth = 1) +
  labs(x="Endemics",
       y="Frequency")
```

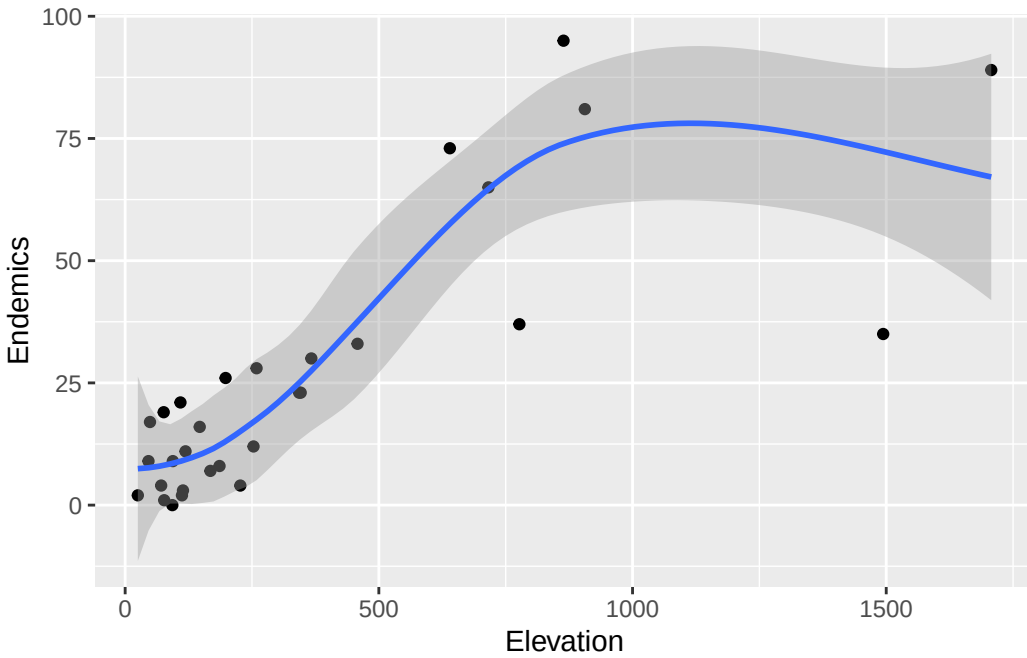


The distribution is skewed to the left. The response variable is constrained to non negative integers.

Q1.4 Plot Endemics ~ Elevation

```
ggplot(gala, aes(x=Elevation, y=Endemics)) +
  geom_point() +
  geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
  labs(x="Elevation",
       y="Endemics")
```

```
<ggplot2::labels> List of 2
 $ x: chr "Elevation"
 $ y: chr "Endemics"
```

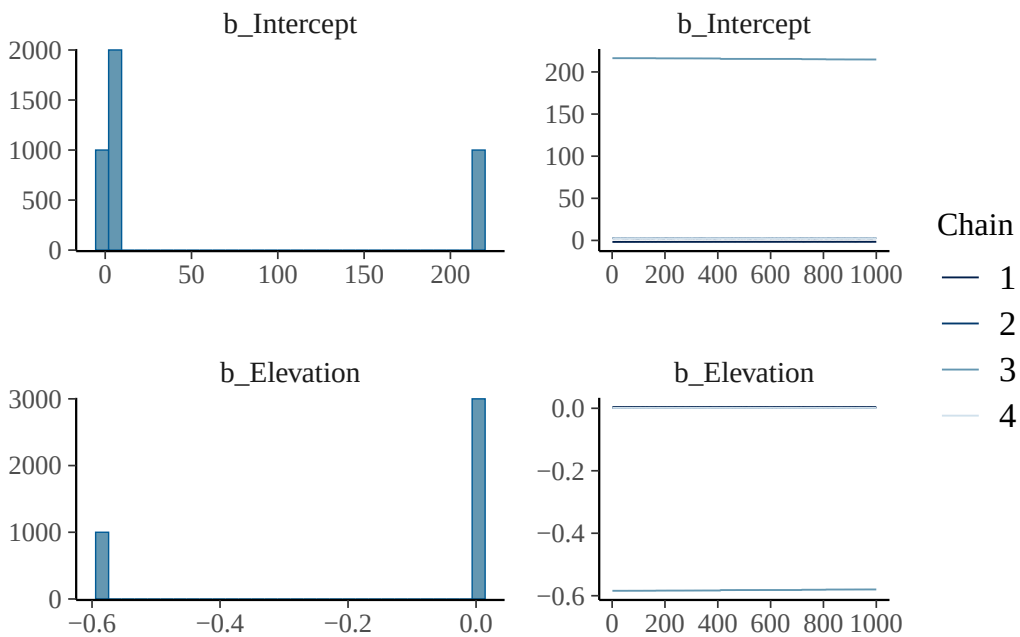
Run the model

```
# Endemics ~ Elevation
m.elev <-
  brm(data = gala, # Give the model the penguins data
      # Choose a poisson distribution - THIS IS THE NEW PART!
      family = poisson(link = "log"),
      # Specify the model here.
      Endemics ~ 1 + Elevation,
```

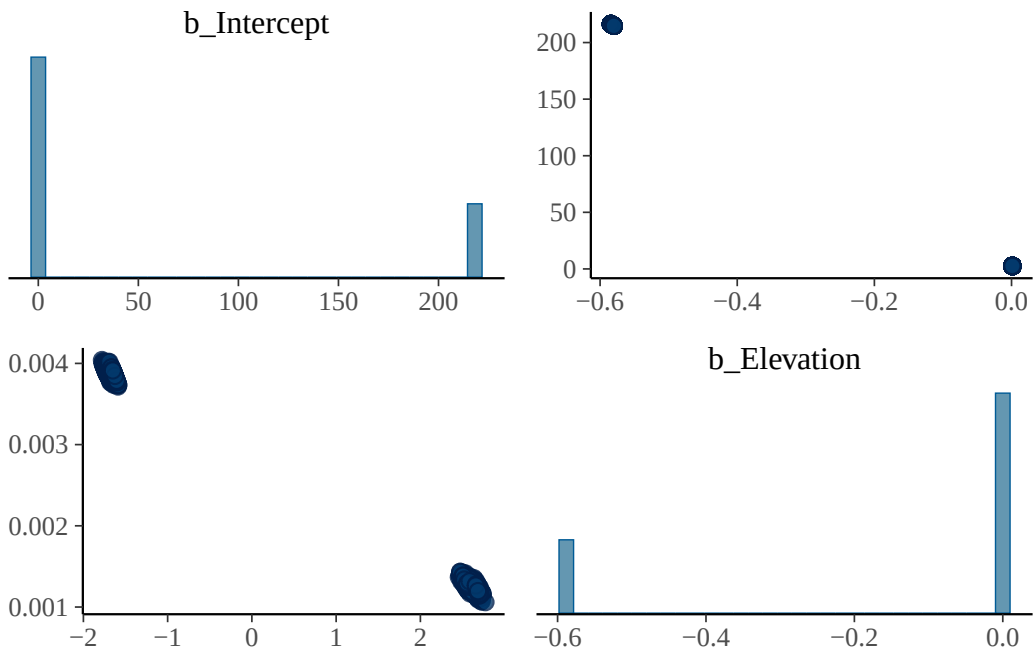
```
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.elev")
```

And let's check out the summary:

```
plot(m.elev)
```




```
pairs(m.elev)
```




```r
summary(m.elev)
```

```
Warning: Parts of the model have not converged (some Rhats are > 1.05). Be
careful when analysing the results! We recommend running more iterations and/or
setting stronger priors.

 Family: poisson
  Links: mu = log
Formula: Endemics ~ 1 + Elevation
   Data: gala (Number of observations: 30)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    54.79     92.87    -1.73   216.31 2.96        5       11
Elevation    -0.14      0.25    -0.58     0.00 2.72        5       11

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Q1.5 Evaluate the output

The model did not run great, rhat is very high with a value of 2.96. The chains do not converge and overlap each other, the posterior distributions are very skewed and has extreme outliers. The CI values are not non zero and overlap with zero.

Q1.6 Center the predictors

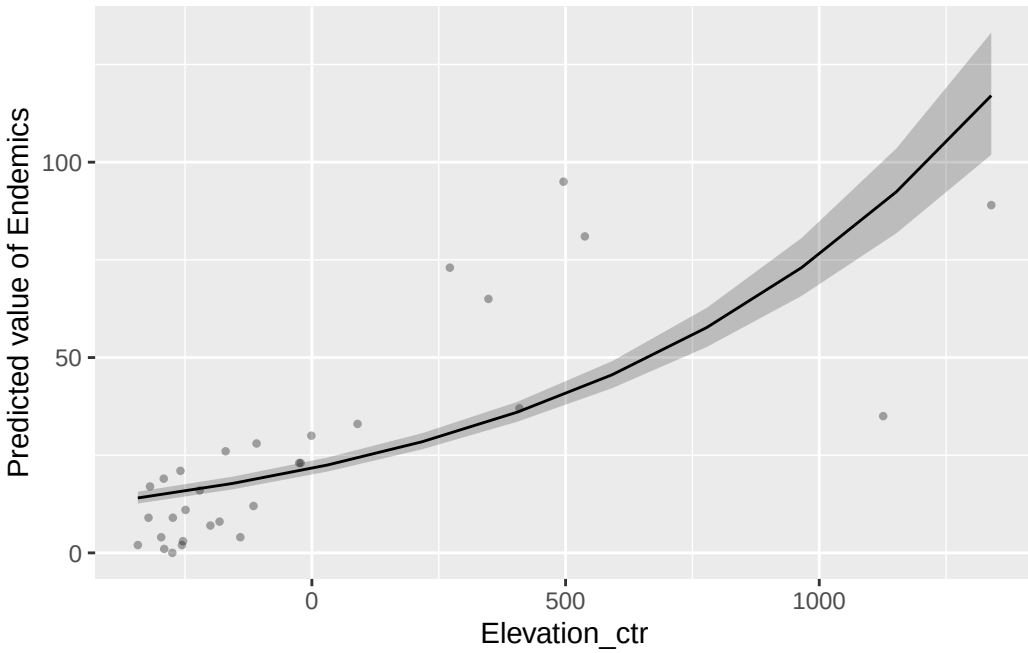
To fix convergence issues, it helps to center the predictors, increase the iterations and warmups, and set some priors. Please 1) center Elevation by creating a new column, `Elevation_ctr`, that subtracts each value in the `Elevation` column by the mean value (hint: `Elevation - mean(Elevation)`), 2) change the brm chain arguments to iter=6000, warmup=4000, 3) add some priors that seem reasonable, and 4) re-fit the model. Please call the new model m.elev2, then use plot(), summary(), and pairs() on it. This is a lot on your own, but you can do it! And we're here to help.

Remember, the model output rounds to two decimal places, so if the values are smaller than two decimal places, you can replace `summary(model)` with `print(model, digits = 4)`.

Plot the posterior

Before we get into interpreting the GLM output, let's plot the posterior prediction on the measurement scale using the `estimate_expectation()` function from the `modelbased` package.

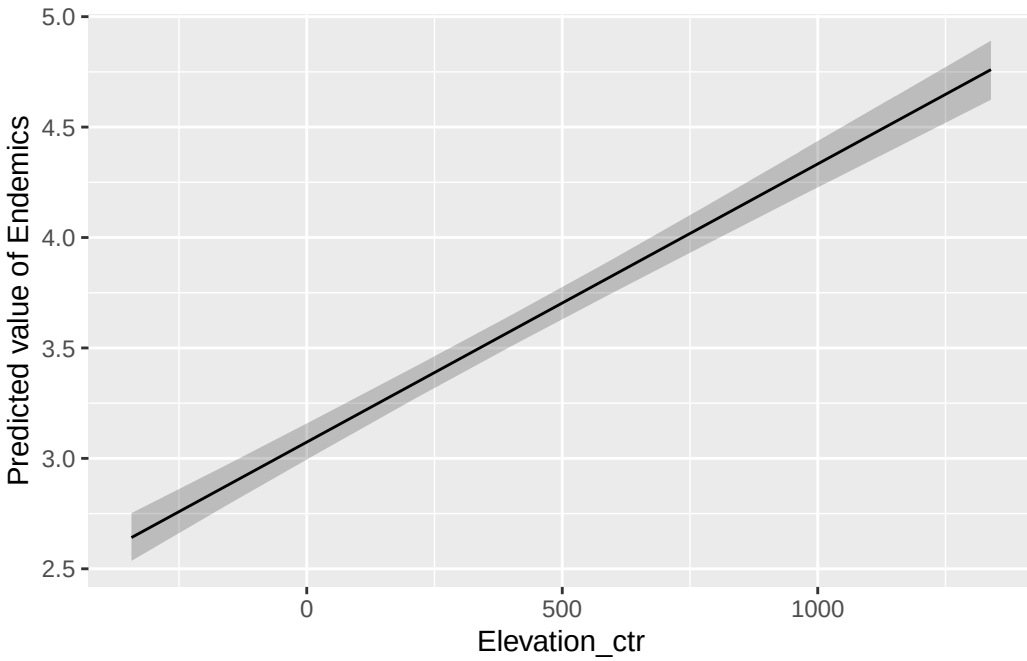
```
preds <- estimate_expectation(m.elev2, by = 'Elevation_ctr')
plot(preds, show_data = TRUE) #posterior plotted, had to install the "see" package
```



Remember how we have a log link? What that means is that the model ran a linear model on the **logged** values of our response, `Endemics`. So the output that we get from the model is reporting the values of our slope, intercept, etc on the log scale.

The `predict_response()` function by default back-transforms from the log scale (log(`Endemics`)) of the linear model to the measurement scale of `Endemics`; that's what you see above. To see the linear model on the log scale, however, we can set predict = 'link':

```
predslog <- estimate_expectation(m.elev2, by = 'Elevation_ctr', predict = 'link')
plot(predslog)
```



Interpreting link scale coefficients

Q1.7 What is the percent change on the response scale?

1. Number of Clarkias blooming as a function of temperature in Celsius: 1.09

For every 1 C degree increase in temperature, Clarkias blooming increases by 197.4%.

```
exp(1.09)
```

```
[1] 2.974274
```

```
(2.974274-1.00)*100
```

```
[1] 197.4274
```

2. Density of sea urchins per square meter in a quadrat as a function of number of sea otters: -2.5

For every additional otter, the density of sea urchins per square meter decreases by 91.8%.

```
exp(-2.5)
```

```
[1] 0.082085
```

```
(0.082085-1.00)*100
```

```
[1] -91.7915
```

3. Number of tomatoes per plant as a function of kg of fertilizer: 6.24

For every kg of fertilizer, there is an 51,185.9% increase of tomatoes per plant.

```
exp(6.24)
```

```
[1] 512.8585
```

```
(512.8585-1.00)*100
```

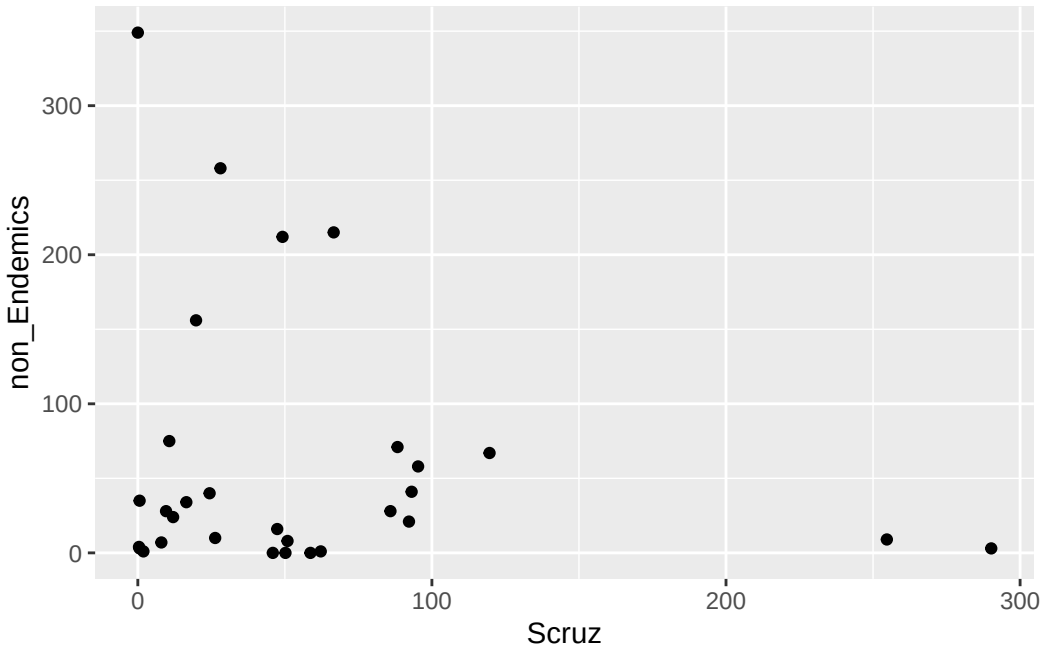
```
[1] 51185.85
```

DIY: Run a model of non-endemic species ~ distance from Santa Cruz Island

Q1.8 Create a non-endemic column

```
gala$non_Endemics <- gala$Species - gala$Endemics #$ pulls out column names from the data in
```

```
ggplot(gala, aes(x=Scruz, y=non_Endemics)) +
  geom_point()
```



Q1.9 Run a model of non Endemics ~ distance from Santa Cruz Island

```
# Endemics \~ distance from Santa Cruz Island
m.edem.dist <-
  brm(data = gala, # Give the model the gala data
      family = poisson(link = "log"), #log function
      non_Endemics ~ 1 + Scruz,
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      file = "output/m.edem.dist")
```

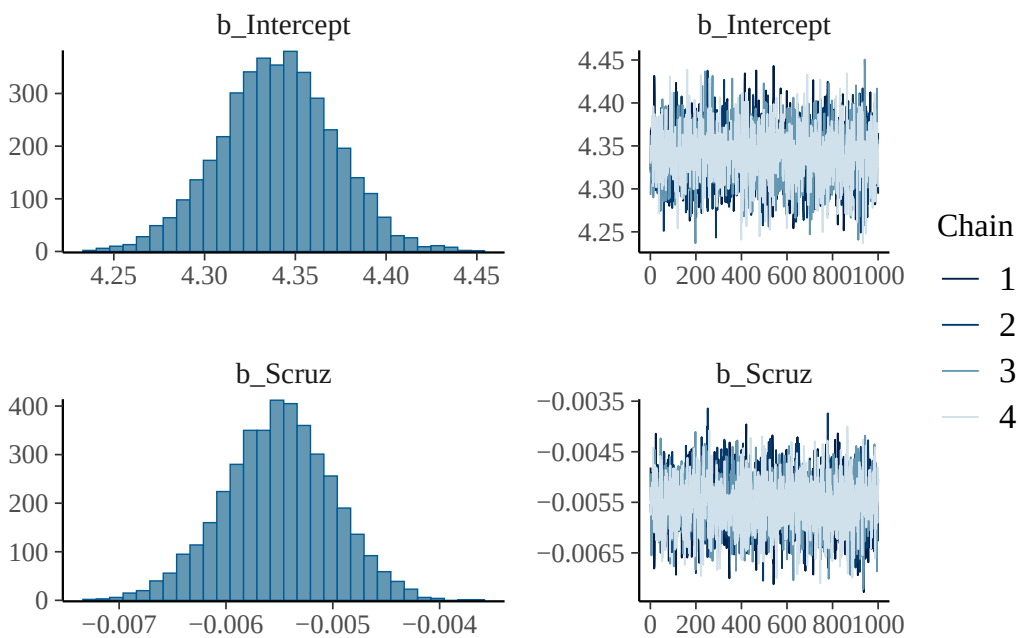
Q1.10 Evaluate the output

```
summary(m.edem.dist)
```

```
 Family: poisson
  Links: mu = log
```

```
Formula: non_Endemics ~ 1 + Scruz
   Data: gala (Number of observations: 30)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept     4.34      0.03     4.28     4.40 1.00     2195     2396
Scruz        -0.01      0.00    -0.01    -0.00 1.00     2904     2511

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m.edem.dist)
```



The model looks like it ran correctly, the chains overlap each other and the posterior distrubtion has one clean peak. Rhat is 1 and the intercept does not overlap zero.

Q1.11 Interpret the output

1. What is the effect of distance from Santa Cruz Island on number of non-endemic species? Report the a) original output on the log scale, b) your backtransformed value, and c) the percent change that this translates to. Describe the effect using the proper units.

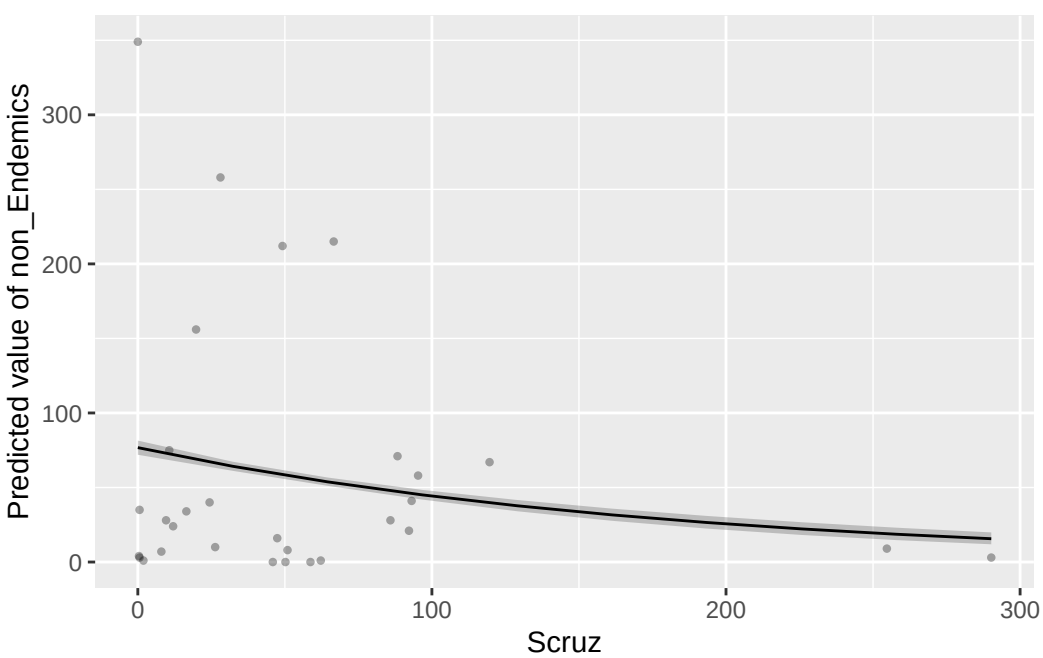
The original log scale slope is -0.01, when backtransformed, the value becomes 0.9900498. Subtracting 1.00 from this value and multiplying by 100 gives us a percentage of -0.99% ~ -1%. This means that for every 1 km increase away from Santa Cruz Island, there is an 1% decrease in non endemic species.

2. Does it seem like the slope estimate is different from zero? Why?

It does not, the 95% CI values are -0.01 and 0.00. We cannot confidently say that the slope is different than zero since these CI values overlap 0.

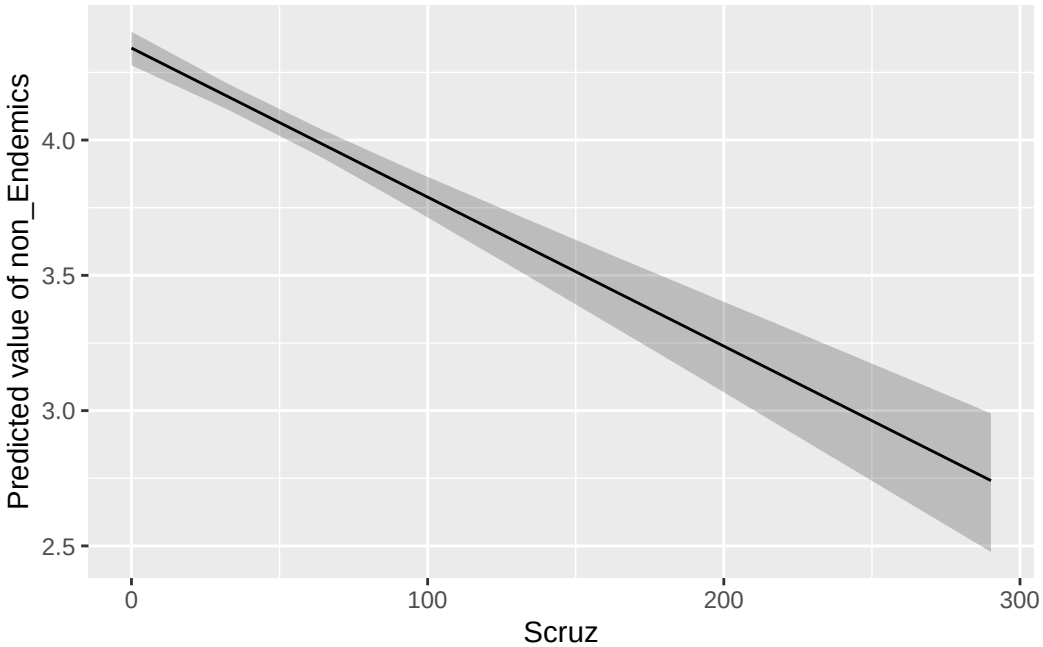
Q1.12 Plot the posterior

```
preds <- estimate_expectation(m.edem.dist, by = 'Scruz')
plot(preds, show_data = TRUE)
```



```
#response scale
```

```
predslog <- estimate_expectation(m.edem.dist, by = 'Scruz', predict = 'link')
plot(predslog)
```




```
#log link scale
```

1.3 GLM with a logit link

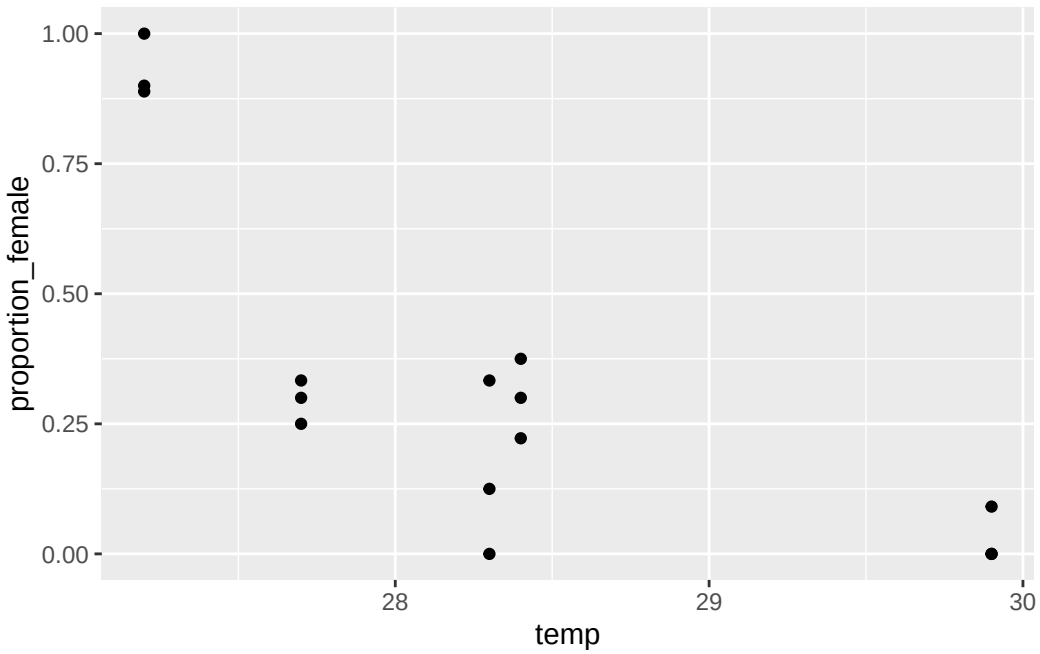
Bring in turtle data

In some species, incubation temperature of eggs affects the sex ratio of males vs females at birth. This dataset, from the `faraway` package, has the number of male and female turtles hatched at different temperatures. Let's calculate the total hatched and the proportion female:

```
turtle <- faraway::turtle %>%
  mutate(total_turtles = male + female,
         proportion_female = female/total_turtles)
```

And let's plot it!

```
turtle %>%
  ggplot(aes(x = temp, y = proportion_female)) +
  geom_point()
```



Run model

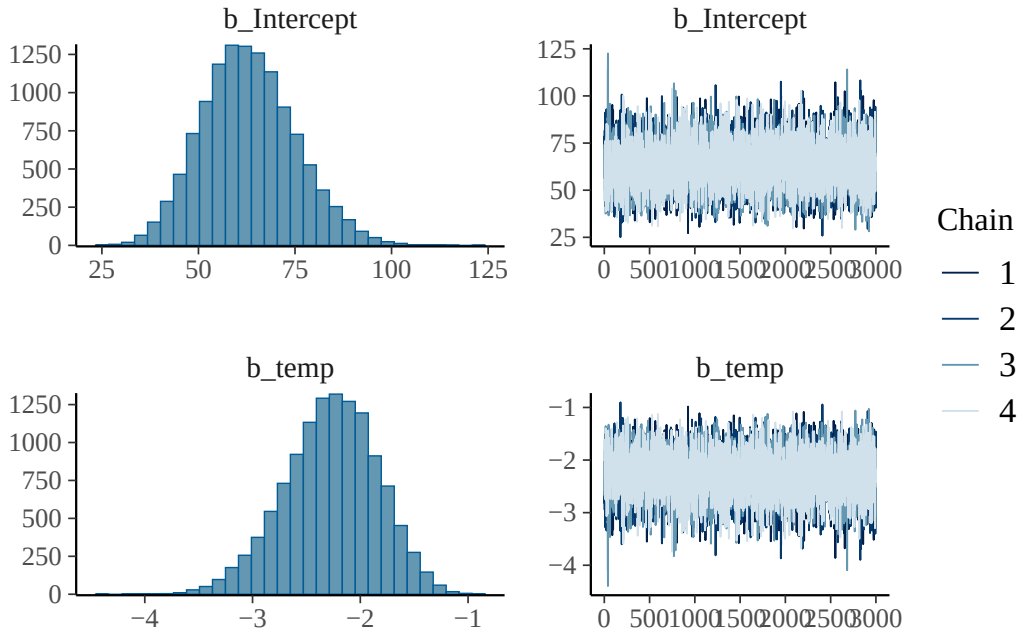
Since these are proportion data, a binomial link is the right choice of distribution to use. Let's fit a model. In this case, we need to specify our integer value of "successes" that we are interested in modeling while also including the number of "trials". In this case, "trials" are the total number of turtles while "successes" are the total number of females.

```
m.turt <-
  brm(data = turtle, # Give the model the data
      # Choose a binomial distribution - THIS IS THE NEW PART!
      family = binomial(link = "logit"),
      # Specify the model here.
      female | trials(total_turtles) ~ 1 + temp,
```

```r
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 4000, warmup = 1000, chains = 4, cores = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.turt")

summary(m.turt)
```

```
 Family: binomial
  Links: mu = logit
Formula: female | trials(total_turtles) ~ 1 + temp
   Data: turtle (Number of observations: 15)
  Draws: 4 chains, each with iter = 4000; warmup = 1000; thin = 1;
         total post-warmup draws = 12000

Regression Coefficients:
          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept    63.05     12.07    40.90    88.36 1.00     5293     5623
temp         -2.27      0.43    -3.18    -1.48 1.00     5230     5595

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
plot(m.turt)
```

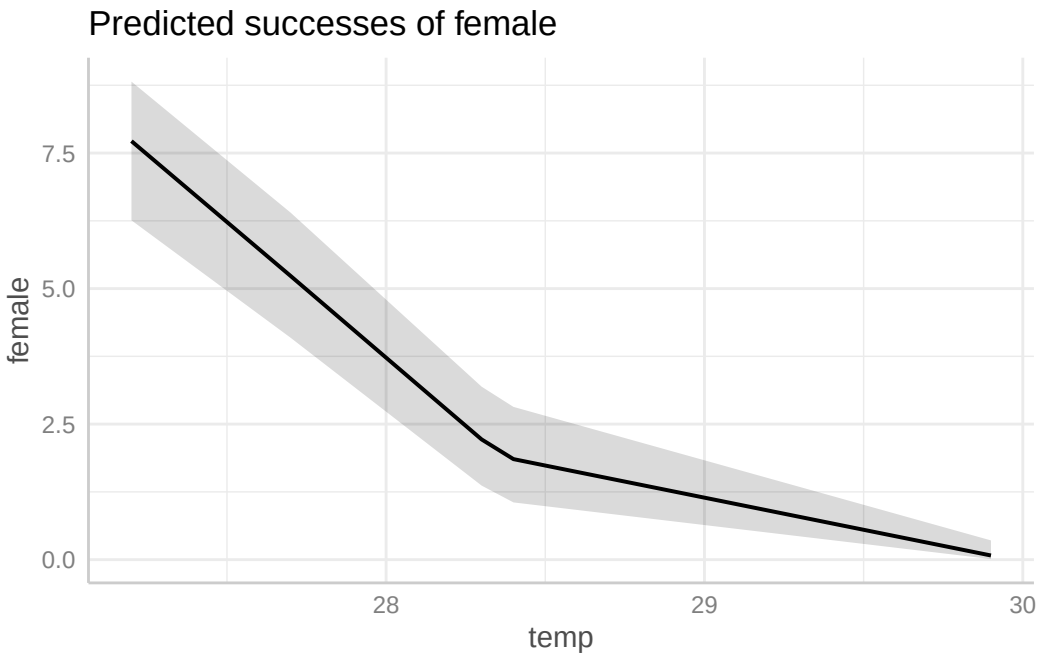



```
#looks like it ran correctly!
```

Interpret model and plot predictions

According to this model, temperature reduces the probability that a turtle will be female by 2.27 on the logit scale per degree Celsius. This is tricky to get into the response scale, so we won't get into that now! Here's some slightly janky code that shows, based on this model, the predicted number of females as a function of temperature. It's standardized at 10 trials, aka 10 total turtles, so a value of 9 means 9 out of 10 turtles are female, and thus there's a 90% probability of being female.y

```
pred <- predict_response(m.turt, condition = c(total_turtles = 10))

plot(pred)
```



2. Multilevel models

Much of the data in ecological and evolutionary biology is clustered in some way. This clustering can be spatial, with some replicate measurements being taken closer to one another than others, such as conducting 10 transects at one site and another 10 transects and a different site. It can temporal, where a bunch of measurements are closer together in time compared to others, such as all within the same season. A single individual, such as with large, well-studied marine mammals, or test subjects in a lab, may be sampled multiple times. There are many other examples!

If we're not careful, we may incorrectly assume that all measurements are independent of one another, where in reality, the measurements that are closer together in space, time, etc, are likely more closely related to one another than to other measurements. Using multi-level models (aka mixed effects models, hierarchical models) can fix this and improve our estimates of our parameters.

For some simple terminology: in multi-level models our predictor variables can be either *fixed effects* or *random effects*

- Fixed effects are the variables we care about getting a parameter estimate from, like a slope value of how x affects y. **Every model we have run so far has used exclusively fixed effects**
- Random effects are the variables that contribute to clustering in our data. They are many coefficients of the same type, and so we can help the model learn about them as a set.
- Sometimes, a variable can be treated as either a random or a fixed effect.

Conceptual practice

Q2.1 Fixed effects vs random effects

1. Student high school graduation rates as a function of: parental income, state of residence, and school district

   Fixed effects: Parental income

   Random effects: State of residence, school district

2. Density of kelp as a function of: latitude, site, transect number, and density of sea urchins

   Fixed effects: Density of sea urchins

   Random effects: Latitude, site, transect number

3. Probability of whale giving birth as a function of: age, annual temperature, year, individual ID

   Fixed effects: annual temperature, age

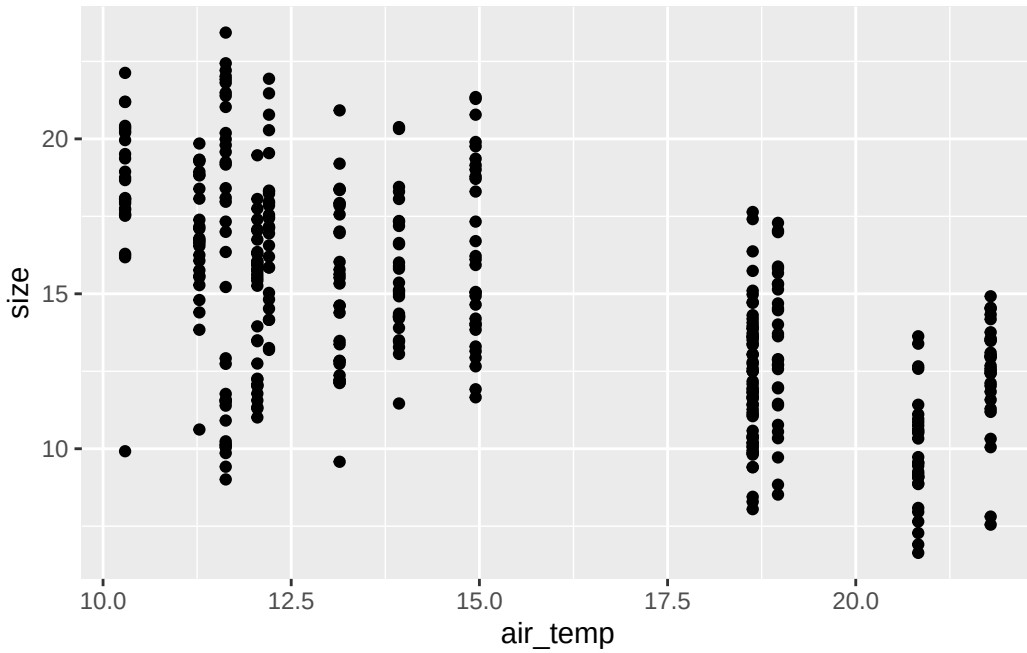
   Random effects: year, individual ID

Incorporate random effects

Let's go back to fiddler crabs and revisit the relationship between `air_temp` and crab `size`:

```
pie_crab <- lterdatasampler::pie_crab %>%
  mutate(site = as.factor(site))
```

```r
pie_crab %>%
  ggplot(aes(x = air_temp, y = size)) +
  geom_point() #relationship between air temp and crab size
```



```r
m.watertemp <-
  brm(data = pie_crab, # Give the model the crab data
      # Use a gamma distribution
      family = Gamma(link = "log"),
      # Specify the model here.
      size ~ 1 + water_temp,
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.watertemp")

print(m.watertemp, digits = 3) #ran well
```

```
 Family: gamma
```

```
  Links: mu = log
Formula: size ~ 1 + water_temp
   Data: pie_crab (Number of observations: 392)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept     3.356     0.055    3.248    3.464 1.000     4213     2946
water_temp   -0.038     0.003   -0.044   -0.032 1.000     4330     2938

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape   23.288     1.664   20.141   26.698 1.002     3032     2664

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Now let's incorporate site as a random effect:

```r
m.watertemp.site <-
  brm(data = pie_crab, # Give the model the penguins data
      # Use a gamma distribution
      family = Gamma(link = "log"),
      # Specify the model here.
      size ~ 1 + water_temp + (1|site),
      # Here's where you specify parameters for executing the Markov chains
      # We're using similar to the defaults, except we set cores to 4 so the analysis runs fa
      iter = 2000, warmup = 1000, chains = 4, cores = 4,
      # Save the fitted model object as output - helpful for reloading in the output later
      file = "output/m.watertemp.site")

print(m.watertemp.site, digits = 3) #ran well
```

```
 Family: gamma
  Links: mu = log
Formula: size ~ 1 + water_temp + (1 | site)
   Data: pie_crab (Number of observations: 392)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000
```

```
Multilevel Hyperparameters:
~site (Number of levels: 13)
              Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
sd(Intercept)    0.126     0.035    0.076    0.210 1.006      939     1272

Regression Coefficients:
           Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
Intercept     3.366     0.194    2.977    3.740 1.003     1175     1515
water_temp   -0.039     0.011   -0.060   -0.018 1.004     1232     1626

Further Distributional Parameters:
      Estimate Est.Error l-95% CI u-95% CI  Rhat Bulk_ESS Tail_ESS
shape   30.089     2.136   26.004   34.357 1.001     3698     2963

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

Q2.2 What is the effect of water_temp on crab size on the response scale?

1. The original output on the log scale is -0.039 when looking at the effect of water temp on crab size. The backtransformed output is 0.96 and the percentage change is -3.82%. This means that for every 1C increase in water temp, there an 3.82% decrease in crab size.

   ```r
   exp(-0.039)
   ```

   ```
   [1] 0.9617507
   ```

   ```r
   (0.9617507-1.00)*100
   ```

   ```
   [1] -3.82493
   ```

2. It does mean like the slope estimate is different from zero. The 95% upper and lower CI values do not overlap zero meaning that we can confidently say that the slope estimate is different than zero.

Q2.3 Compare WAIC and PSIS of the two models

The model including site has better predictive power, it has lower WAIC and PSIS values of the two models.

```r
loo(m.watertemp.site)
```

```
Computed from 4000 by 392 log-likelihood matrix.

         Estimate   SE
elpd_loo   -938.5 13.7
p_loo        12.3  0.8
looic      1877.1 27.4
------
MCSE of elpd_loo is 0.1.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.7, 1.8]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

```r
waic(m.watertemp.site)
```

```
Computed from 4000 by 392 log-likelihood matrix.

          Estimate   SE
elpd_waic   -938.5 13.7
p_waic        12.2  0.8
waic        1877.0 27.4
```

```r
loo(m.watertemp)
```

```
Computed from 4000 by 392 log-likelihood matrix.

         Estimate   SE
elpd_loo   -984.7 12.2
p_loo         2.8  0.2
looic      1969.4 24.3
------
```

```
MCSE of elpd_loo is 0.0.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.8, 1.1]).

All Pareto k estimates are good (k < 0.7).
See help('pareto-k-diagnostic') for details.
```

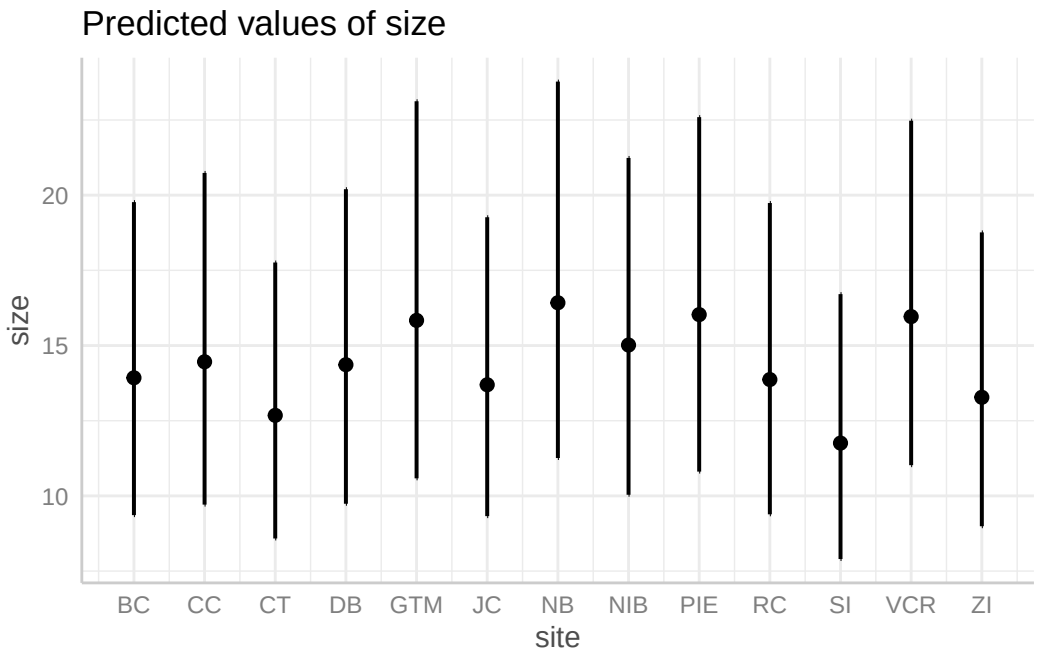
```r
waic(m.watertemp)
```

```
Computed from 4000 by 392 log-likelihood matrix.

          Estimate   SE
elpd_waic   -984.7 12.2
p_waic         2.7  0.2
waic        1969.4 24.3
```

Predict response of random effects

From the lecture, here's some code about how to show variation around each level of each random effect:

```r
preds <- predict_response(m.watertemp.site,
                          interval = "prediction",
                          terms = "site",
                          type = "random")

plot(preds)
```


Predicted values of size
size
20
15
10
BC
CC
CT
DB
GTM
JC
NB
NIB
PIE
RC
SI
VCR
ZI
site